

Binafsha Bakhramova

Boston, MA (Willing to Relocate) | bakhramova.b@northeastern.edu | (347) 893-4988

[LinkedIn](#) • [GitHub](#)

EDUCATION

Northeastern University, D'Amore-McKim School of Business

BS in Business Administration and Political Science, Minor in Data Science

Concentrations: Business Analytics, FinTech, Public Policy

Boston, MA

Expected May 2029

GPA: 3.7

- **Honors & Activities:** Johns Martinson Honors Program, Dean's List, Women in Business; featured guest on NBC Townhouse discussing AI technology (June 2024)
- **Relevant Coursework:** Foundations of Data Science, Business Statistics, Financial Accounting for Advisors and Managers, International Relations

RESEARCH EXPERIENCE

Computational Redistricting Research | [Live Demo](#) | [GitHub](#)

Independent Research

Boston, MA

2026 – Present

- Built a redistricting analysis in Python with GerryChain (ReCom), generating an MCMC ensemble of 500 valid, population-balanced plans to test whether an enacted map is a statistical outlier against neutrally drawn alternatives.
- Found Texas's 2025 enacted map yields 8 Democratic seats while none of the 500 neutral maps fell below 10, placing it at the 0th percentile of the ensemble.

Northeastern University — Data Science Research Project | [Live Site](#)

Machine Learning Developer

Leuven, Belgium

May 2026 – Present

- Built a 2,354-row EU labor panel (27 countries, 8 sectors, 2012–2023) from 2 Eurostat datasets, hand-engineering an ISCED-NACE crosswalk to map education fields to industry sectors and adding lag and year-over-year features.
- Caught that a near-perfect R^2 came from the lag feature dominating rather than real predictive skill, and switched to a temporal train/test split (train 2012–2020, test 2021–2023) to eliminate time-series leakage.
- Productionized the models into a Flask backend storing model weights in MySQL parameter tables, then shipped a Streamlit dashboard with Plotly charts for scenario inspection across countries and sectors.
- Built a budget-reallocation model (per-sector OLS employment trend plus graduate absorption rate, z-scored and softmaxed into budget shares) served through a JSON route, flagging CS as underfunded (+\$595K) and Engineering as overfunded (–\$655K) for Belgium.

Boston Area Research Initiative (BARI)

Research Assistant

Boston, MA

January 2026 – July 2026

- Built a real-time public heat-alert map for the Blue Hill Ave corridor (JavaScript, Open-Meteo API, Leaflet.js) turning live sensor readings into accessible visuals, contributing to data-pipeline and tool design for the Common SENSES sensing project.

NYU Tandon School of Engineering

Research Assistant

New York, NY

June 2024 – August 2024

- Benchmarked 3 control strategies for precision-robotics tracking by simulating NYU-Finger robot dynamics in PyBullet, then presented the most accurate, energy-efficient controller in a thesis-style defense.

TECHNICAL SKILLS

Languages: Python, R, JavaScript

Quantitative & Data: statistical modeling, regression, Monte Carlo simulation, model validation, time-series, Pandas, NumPy, scikit-learn

Tools & Frameworks: Flask, Streamlit, GerryChain, geopandas, MySQL, Git/GitHub, Plotly, REST APIs, Excel

Languages (Spoken): English (Native), Russian (Professional Working Proficiency)

LEADERSHIP & ACTIVITIES

Northeastern University AI Advisory Board

Member

Boston, MA

April 2026 – Present

- Advising university leadership on institutional AI strategy across 3 focus areas, partnering with academic centers to expand student AI access and redesign classroom integration across Northeastern's 9 colleges.

EVH Hall Council

Vice President (Budget)

Boston, MA

February 2026 – April 2026

- Tracked and reported a \$7,500 budget (13% of EVH funding), monitoring spending against plan and preparing financial documentation for review and approval.

Accelerate Pre-College Programs

Student Life Ambassador

Boston, MA

June 2026 – Present

- Mentored 100+ first-time attendees in a 600-student pre-college program and helped run its community events.